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CONTROL DEVICE, SYSTEM, MOUNTING DEVICE, AND MOVING OBJECT

Abstract

A control device controls mounting of a component on a moving object in a process of manufacturing the moving object, and comprises: an information acquiring unit that acquires first information that is information relating to unsuccessful mounting of the component by a mounting device to conduct on-the-move mounting that is mounting of the component on the moving object while the moving object is moving by unmanned driving; a first instructing unit that gives a first instruction for instructing deceleration or stop to the moving object if the first information is acquired; and a second instructing unit that gives a second instruction for instructing re-mounting to the mounting device after transmission of the first instruction. The re-mounting is mounting of the component on the moving object while the moving object is decelerated or stopped.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
[0001] The present application claims priority from Japanese patent application No. P2024-027141 filed on Feb. 27, 2024, the disclosure of which is hereby incorporated in its entirety by reference into the present application.
BACKGROUND
Field
[0002] The present disclosure relates to a control device, a system, a mounting device, and a moving object.
Related Art
[0003] There is a known technique by which a vehicle is driven in a driverless manner by remote control in a process of manufacturing the vehicle (Japanese Patent Application Publication (Translation of PCT Application) No. 2017-538619, for example).
[0004] In the process of manufacturing the vehicle, employing on-the-move mounting is examined by which a component is mounted on the vehicle in a moving state by controlling a robot. However, the above Patent Application Publication No. 2017-538619 does not give sufficient consideration to how to handle a situation in the event of failure of such on-the-move mounting. This problem is not only in the case of vehicles but is common to any type of moving object.
SUMMARY
[0005] According to one aspect of the present disclosure, a control device is provided that controls mounting of a component on a moving object in a process of manufacturing the moving object. The control device comprises: an information acquiring unit that acquires first information that is information relating to unsuccessful mounting of the component by a mounting device to conduct on-the-move mounting that is mounting of the component on the moving object while the moving object is moving by unmanned driving; a first instructing unit that gives a first instruction for instructing deceleration or stop to the moving object if the first information is acquired; and a second instructing unit that gives a second instruction for instructing re-mounting to the mounting device after transmission of the first instruction. The re-mounting is mounting of the component on the moving object while the moving object is decelerated or stopped.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] FIG. **1** is an explanatory view showing the configuration of a system according to a first embodiment;
[0007] FIG. **2** is an explanatory view showing the configuration of a vehicle according to the first embodiment;
[0008] FIG. **3** is an explanatory view showing the configuration of a server device according to the first embodiment;
[0009] FIG. **4** is an explanatory view showing the configuration of a mounting robot according to the first embodiment;
[0010] FIG. **5** is an explanatory view showing how the vehicle runs by unmanned driving in a factory;

[0011] FIG. **6** is a flowchart showing a processing procedure of running control over a vehicle **100** according to the first embodiment;
[0012] FIGS. **7** and **8** are flowcharts each showing a procedure of component mounting control according to the first embodiment;
[0013] FIG. **9** is an explanatory view showing the configuration of a mounting robot according to a second embodiment;
[0014] FIG. **10** is a block diagram showing the configuration of a system according to a third embodiment; and
[0015] FIG. **11** is a flowchart showing a processing procedure of running control over a vehicle according to the third embodiment.

DETAILED DESCRIPTION

A. First Embodiment

A-1. System Configuration:

[0016] FIG. **1** is an explanatory view showing the configuration of a system **10** according to the first embodiment. System **10** is used in a factory KJ for manufacture of a vehicle **100** as a moving object, for example. In the present disclosure, the "moving object" means an object capable of moving, and is a vehicle or an electric vertical takeoff and landing aircraft (so-called flying-automobile), for example. The vehicle may be a vehicle to run with a wheel or may be a vehicle to run with a continuous track, and may be a passenger car, a truck, a bus, a two-wheel vehicle, a four-wheel vehicle, or a construction vehicle, for example. The vehicle includes a battery electric vehicle (BEV), a gasoline automobile, a hybrid automobile, and a fuel cell automobile. When the moving object is other than a vehicle, the term "vehicle" or "car" in the present disclosure is replaceable with a "moving object" as appropriate, and the term "run" is replaceable with "move" as appropriate.

[0017] System **10** includes a server device **200**, at least one external sensor **250**, and a mounting robot **300**. The system **10** controls mounting of a component on the vehicle **100** in the factory KJ for manufacture of the vehicle **100**. The vehicle **100** is configured to be capable of running by unmanned driving. Even when the vehicle **100** is being manufactured in a process of manufacturing the vehicle **100** itself, the vehicle **100** is still capable of running by unmanned driving. In the system **10**, the mounting robot **300** conducts the work of mounting a component on the vehicle **100** running by unmanned driving. In the present embodiment, the vehicle **100** runs by unmanned driving while the vehicle **100** is in the state of a so-called platform.

[0018] To fulfill three functions including "run," "turn," and "stop" by unmanned driving, the vehicle **100** in the state of a platform may be required to include at least a control device for controlling running of the vehicle **100**, and actuators such as a driving device, a steering device, and a braking device. If the vehicle **100** is to acquire information from outside for unmanned driving, the vehicle **100** may further include a communication device. Specifically, the vehicle **100** movable by unmanned driving may not have to be equipped with at least some of interior components such as a driver's seat and a dashboard, may not have to be equipped with at least some of exterior components such as a bumper and a fender, or may not have to be equipped with a bodyshell. In such cases, a remaining component such as a bodyshell may be mounted on the vehicle **100** before the vehicle **100** is shipped from the factory KJ, or a remaining component such as a bodyshell may be mounted on the vehicle **100** after the vehicle **100** is shipped from the factory KJ while the remaining component such as a bodyshell is not mounted on the vehicle **100**. Each of components may be mounted on the vehicle **100** from any direction such as from above, from below, from the front, from the rear, from the right, or from the left. These components may be mounted from the same direction or from respective different directions.

[0019] In the present embodiment, "unmanned driving" means driving independent of driving operation by a passenger on-board the vehicle **100**. "Driving operation" means operation relating to at least one of "run," "turn," and "stop" of the vehicle **100**. Unmanned driving is realized by automatic or manual remote control using a device provided outside the vehicle **100**, or by autonomous control by the vehicle **100**. A passenger not involved in driving operation may be on-board the vehicle **100** running by unmanned driving. The passenger not involved in driving operation includes a person simply sitting in a driver's seat of the vehicle **100** and a person doing behavior different from driving operation, for example. The behavior different from driving operation includes work of mounting a component on the vehicle **100**, inspection on the vehicle **100**, and operation of switches provided in the vehicle **100**, for example. Driving by driving operation by a passenger may be called “manned driving.”

[0020] The vehicle **100** is configured to be capable of running by unmanned driving. The “unmanned driving” means driving independent of running operation by a passenger. The running operation means operation relating to at least one of “run,” “turn,” and “stop” of the vehicle **100**. Unmanned driving is realized by automatic remote control or manual remote control using a device provided outside the vehicle **100** or by autonomous control by the vehicle **100**. A passenger not involved in running operation may be on-board a vehicle running by the unmanned driving. The passenger not involved in running operation includes a person simply sitting in a seat of the vehicle **100** and a person doing work such as assembly, inspection, or operation of switches different from running operation while on-board the vehicle **100**. Driving by running operation by a passenger may also be called “manned driving.” In the present specification, the “remote control” includes “complete remote control” by which all motions of the vehicle **100** are completely determined from outside the vehicle **100**, and “partial remote control” by which some of the motions of the vehicle **100** are determined from outside the vehicle **100**. The “autonomous control” includes “complete autonomous control” by which the vehicle **100** controls a motion of the vehicle **100** autonomously without receiving any information from a device outside the vehicle **100**, and “partial autonomous control” by which the vehicle **100** controls a motion of the vehicle **100** autonomously using information received from a device outside the vehicle **100**.

[0021] A vehicle control device **110** is configured using a computer including a processor **111**, a memory **112**, an input/output interface **113**, and an internal bus **114**. The processor **111**, the memory **112**, and the input/output interface **113** are connected to each other via the internal bus **114** in a manner allowing bidirectional communication therebetween. An actuator group **120** and a communication device **130** are connected to the input/output interface **113**. The processor **111** executes a program PG**1** stored in memory **112**, thereby realizing various functions including a function as a vehicle controller **115**.

[0022] The vehicle controller **115** causes the vehicle **100** to run by controlling the actuator group **120**. The vehicle controller **115** is capable of causing the vehicle **100** to run by controlling the actuator group **120** using a running control signal received from the server device **200**. The running control signal is a control signal for causing the vehicle **100** to run. In the present embodiment, the running control signal includes the acceleration and rudder angle of the vehicle **100** as parameters. In other embodiments, the running control signal may include the speed of the vehicle **100** as a parameter instead of or in addition to the acceleration of the vehicle **100**. When a passenger is on-board the vehicle **100**, the vehicle controller **115** is capable of causing the vehicle **100** to run by controlling the actuator group **120** in response to driving operation by the passenger. Independently of whether a passenger is on-board the vehicle **100**, the vehicle controller **115** is capable of causing the vehicle **100** to run by controlling the actuator group **120** in response to a running control signal received from the server device **200**. The vehicle controller **115** corresponds to a “moving object controller” of the present disclosure.

[0023] FIG. **3** is an explanatory view showing the configuration of the server device **200** according to the present embodiment. The server device **200** is configured using a computer including a processor **201**, a memory **202**, an input/output interface **203**, and an internal bus **204**. Processor **201**, the memory **202**, and the input/output interface **203** are connected to each other via the internal bus **204** in a manner allowing bidirectional communication therebetween. A communication device **205** for communicating with the vehicle **100** by radio communication is connected to the input/output interface **203**. In the present embodiment, communication device **205** is capable of communicating with the external sensor **250** and the mounting robot **300** via wire communication or radio communication. The processor **201** executes a program PG**2** stored in advance in the memory **202**, thereby functioning as a calculating unit **210**, a vehicle control commanding unit **212**, an error information acquiring unit **214**, a stop instructing unit **216**, and a robot control commanding unit **218**. The server device **200** corresponds to a "control device" of the present disclosure.

[0024] In the present embodiment, the calculating unit **210** calculates vehicle positional information using detection result output from the external sensor **250**. The vehicle positional information contains information about the position and direction of the vehicle **100**. The vehicle control commanding unit **212** generates a vehicle control command using the vehicle positional information. The vehicle control commanding unit **212** generates a running control signal mentioned above as the vehicle control command.

[0025] The error information acquiring unit **214** acquires mounting error information, and switches the state of a stop flag SF stored in memory **202** to either ON or OFF in response to the mounting error information. The "mounting error information" means information about mounting error. The "mounting error" means a state where, even after the mounting robot **300** makes motion of mounting a component, the component is not mounted properly on the vehicle **100**. Specifically, the mounting error information means information indicating that the component has not been mounted successfully on the vehicle **100** by the mounting robot **300**. In the present embodiment, the error information acquiring unit **214** acquires an error signal as the mounting error information output from the mounting robot **300**, as will be described later. The "stop flag SF" means a flag representing a situation of the occurrence of mounting error in the system **10**. A way in which the stop flag SF is switched will be described later in detail. The mounting error information corresponds to "first information" of the present disclosure. The error information acquiring unit **214** corresponds to an "information acquiring unit" of the present disclosure.

[0026] The stop instructing unit **216** transmits a stop signal for instructing stop to the vehicle **100** if the mounting error information is acquired. In the present embodiment, the stop signal is an emergency stop signal for instructing the vehicle **100** to be stopped emergently and may be generated independently of detection result from the external sensor **250**. This allows the stop instructing unit **216** to instruct emergent stop of the vehicle **100** immediately in response to acquisition of the mounting error information independently of detection result from the external sensor **250**. If the stop signal is received, the vehicle controller **115** controls the actuator group **120** to stop the vehicle **100**. Specifically, the stop signal corresponds to a "first instruction" of the present disclosure and the stop instructing unit **216** corresponds to a "first instructing unit" of the present disclosure.

[0027] The robot control commanding unit **218** generates a motion control signal for causing the mounting robot **300** to make motion and transmits the motion control signal to the mounting robot **300**. The mounting robot **300** having received the motion control signal, makes motion in response to the motion control signal. In the following description, the motion control signal for instructing the mounting robot **300** to make motion of mounting a component on the vehicle **100** will also be called a "mounting instruction." The motion control signal of the present embodiment is generated as a signal for making a specific command for a stroke of each part forming the mounting robot **300**. The robot control commanding unit **218** of the present embodiment corresponds to a "motion control signal generating unit."

[0028] In the present embodiment, the robot control commanding unit **218** generates the motion control signal for the mounting robot **300** in such a manner as to cause the mounting robot **300** to conduct mounting work while the vehicle **100** is in a state of moving by unmanned driving. In the following description, such mounting of a component on the vehicle **100** moving by unmanned driving will also be called "on-the-move mounting." According to the on-the-move mounting, as work is conducted while the vehicle **100** is running, it is possible to prevent loss in manufacturing speed during implementation of the work. The robot control commanding unit **218** generates the motion control signal in such a manner as to mount a component on a predetermined position of the vehicle **100** in a predetermined direction while the vehicle **100** is running and to achieve a zero relative speed during the mounting between the vehicle **100** and the component in a direction parallel to a mounting target surface of the vehicle **100**.

[0029] The external sensor **250** is located outside the vehicle **100**. The external sensor **250** is used for detecting the position and direction of the vehicle **100**. In the present embodiment, the external sensor **250** is a camera installed at the factory KJ. The external sensor **250** includes a communication device not shown in the drawings and is capable of communicating with the server device **200** via wire communication or radio communication. The external sensor **250** is not limited to the camera but may be LiDAR, for example.

[0030] FIG. **4** is an explanatory view showing the configuration of the mounting robot **300** according to the present embodiment. The mounting robot **300** includes a robot control device **310**, an arm part **320**, and a communication device **330**. In the present embodiment, the robot control device **310** controls each part of the mounting robot **300**. The arm part **320** is composed of a vertical articulated type robot arm. The arm part **320** has a tip equipped with an end effector for gripping a component. In the present embodiment, the end effector is configured to put a component therein. The communication device **330** is capable of communicating with the server device **200** via wire communication or radio communication. The arm part **320** is not limited to a vertical articulated type robot arm but may be composed of a horizontal articulated type robot arm, an orthogonal type robot arm, or a parallel link type robot arm, for example. The end effector may be configured to cause a component to stick thereto instead of being configured to put the component therein. While not shown in the drawings, the mounting robot **300** includes sensors for detecting a motor load for driving the arm part **320**, impact force applied during component mounting, etc. The mounting robot **300** corresponds to a "mounting device" of the present disclosure.

[0031] The robot control device **310** is configured using a computer including a processor **311**, a memory **312**, an input/output interface **313**, and an internal bus **314**. The processor **311**, the memory **312**, and the input/output interface **313** are connected to each other via the internal bus **314** in a manner allowing bidirectional communication therebetween. The arm part **320** and the communication device **330** are connected to the input/output interface **313**.

[0032] In the present embodiment, the processor **311** executes a program PG**3** stored in advance in memory **312**, thereby functioning as a robot controller **315**. The robot controller **315** receives a motion control signal and controls each part of the mounting robot **300** including the arm part **320** in response to the motion control signal. The robot controller **315** corresponds to a "receiving unit" and a "device controller" of the present disclosure.

[0033] The robot controller **315** outputs an error signal on the occurrence of error in mounting a component on the vehicle **100**. The mounting error may occur in a case where a component mounting position is not set properly or error is generated in detecting the position of the vehicle **100**, for example. If a motor load or impact force applied during component mounting is not within a threshold range defined in advance, for example, the robot controller **315** determines that the mounting error has occurred and outputs the error signal.

[0034] FIG. **5** is an explanatory view showing how the vehicle **100** runs by unmanned driving in the factory KJ. In the present embodiment, the factory KJ has a first place PL**1**, a second place PL**2**, and a third place PL**3**. The first place PL**1**, the second place PL**2**, and the third place PL**3** are connected to each other via a track SR along which the vehicle **100** is capable of running. The factory KJ is equipped with a plurality of the external sensors **250** along the track SR.

[0035] The first place PL**1** is a place where work of assembling the vehicle **100** is

[0036] conducted. The vehicle **100** having been assembled at the first place PL**1** becomes capable of running by unmanned driving, in other words, becomes capable of fulfilling the three functions including "run," "turn," and "stop" by unmanned driving. In the present embodiment, the vehicle **100** having been assembled at the first place PL**1** is in the state of a platform. The vehicle **100** moves from the first place PL**1** to the second place PL**2** by unmanned driving.

[0037] The second place PL**2** is a place where further work of mounting a component on the vehicle **100** is conducted. The mounting robot **300** is located in the second place PL**2**. A component to be mounted at the second place PL**2** is a vehicle body component, an interior component such as a seat, a headlamp, or a wiper, for example. In the present embodiment, the vehicle **100** with the component mounted at the second place PL**2** is in the state of a finished vehicle. The vehicle **100** moves from the second place PL**2** to the third place PL**3** by unmanned driving.

[0038] The third place PL**3** is a place where work of inspecting the vehicle **100** is conducted. The vehicle **100** having passed the inspection at the third place PL**3** is shipped from the factory KJ. The vehicle **100** is not required to be in the state of a finished vehicle when shipped from the factory KJ. Specifically, there may be a component yet to be mounted on the vehicle **100** when the vehicle **100** is shipped from the factory KJ. In this case, the component yet to be mounted may be mounted on the vehicle **100** after the vehicle **100** is shipped from the factory KJ.

A-2. Running Control:

[0039] FIG. **6** is a flowchart showing a processing procedure for running control of the vehicle **100** in the first embodiment. In step S**1**, the calculating unit **210** acquires vehicle location information of the vehicle **100** using detection result output from an external sensor **250**. The external sensor is located outside the vehicle **100**. The vehicle location information is locational information as a basis for generating a running control signal. In the present embodiment, the vehicle location information includes the location and orientation of the vehicle **100** in a global coordinate system GA of the factory KJ. Specifically, in step S**1**, the calculating unit **210** acquires the vehicle location information using the captured image acquired from the camera as the external sensor **250**.

[0040] More specifically, in step S**1**, the calculating unit **210** for example, determines the outer shape of the vehicle **100** from the captured image, calculates the coordinates of a positioning point of the vehicle **100** in a coordinate system of the captured image, namely, in a local coordinate system, and converts the calculated coordinates to coordinates in the global coordinate system GA, thereby acquiring the location of the vehicle **100**. The outer shape of the vehicle **100** in the captured image may be detected by inputting the captured image DM to a detection model using artificial intelligence, for example. The detection model DM is prepared in the system **10** or outside the system **10**. The detection model is stored in advance in a memory **202** of the server **200**, for example. An example of the detection model DM is a learned machine learning model that was learned so as to realize either semantic segmentation or instance segmentation. For example, a convolution neural network (CNN) learned through supervised learning using a learning dataset is applicable as this machine learning model. The learning dataset contains a plurality of training images including the vehicle **100**, and a label showing whether each region in the training image is a region indicating the vehicle **100** or a region indicating a subject other than the vehicle **100**, for example. In training the CNN, a parameter for the CNN is preferably updated through backpropagation in such a manner as to reduce error between output result obtained by the detection model and the label. The calculating unit **210** can acquire the orientation of the vehicle **100** through estimation based on the direction of a motion vector of the vehicle **100** detected from change in location of a feature point of the vehicle **100** between frames of the captured images using optical flow process, for example.

[0041] In step S**2**, the vehicle control commanding unit **212** determines a target location to which the vehicle **100** is to move next. In the present embodiment, the target location is expressed by X, Y, and Z coordinates in the global coordinate system GA. The memory **202** of the server **200** contains a reference route RR stored in advance as a route along which the vehicle **100** is to run.

The route is expressed by a node indicating a departure place, a node indicating a way point, a node indicating a destination, and a link connecting nodes to each other. The vehicle control commanding unit **212** determines the target location to which the vehicle **100** is to move next using the vehicle location information and the reference route RR. The vehicle control commanding unit **212** determines the target location on the reference route RR ahead of a current location of the vehicle **100**.

[0042] In step S**3**, the vehicle control commanding unit **212** generates a running control signal for causing the vehicle **100** to run toward the determined target location. The vehicle control commanding unit **212** calculates a running speed of the vehicle **100** from transition of the location of the vehicle **100** and makes comparison between the calculated running speed and a target speed of the vehicle **100** determined in advance. If the running speed is lower than the target speed, the vehicle control commanding unit **212** generally determines an acceleration in such a manner as to accelerate the vehicle **100**. If the running speed is higher than the target speed as, the vehicle control commanding unit **212** generally determines an acceleration in such a manner as to decelerate the vehicle **100**. If the vehicle **100** is on the reference route RR, the vehicle control commanding unit **212** determines a steering angle and an acceleration in such a manner as to prevent the vehicle **100** from deviating from the reference route. If the vehicle **100** is not on the reference route RR, in other words, if the vehicle **100** deviates from the reference route RR, the server **200** determines a steering angle and an acceleration in such a manner as to return the vehicle **100** to the reference route.

[0043] In step S**4**, the vehicle control commanding unit **212** transmits the generated running control signal to the vehicle **100**. The vehicle control commanding unit **212** repeats the acquisition of vehicle location information, the determination of a target location, the generation of a running control signal, the transmission of the running control signal, and others in a predetermined cycle.

[0044] In step S**5**, the vehicle controller **115** receives the running control signal transmitted from the server **200**. In step S**6**, the vehicle controller **115** controls an actuator group **120** using the received running control signal, thereby causing the vehicle **100** to run at the acceleration and the steering angle indicated by the running control signal. The vehicle controller **115** repeats the reception of a running control signal and the control over the actuator group **120** in a predetermined cycle. According to the system **10** in the present embodiment, it becomes possible to move the vehicle **100** without using a transport unit such as a crane or a conveyor.

A-3. Component Mounting Control:

[0045] FIGS. **7** and **8** are flowcharts each showing a procedure of component mounting control according to the first embodiment. In the present embodiment, the above-described running control is performed as basic control and the control mentioned herein is performed in combination with this running control. The control shown in FIG. **7** and the control shown in FIG. **8** are started when the system **10** is brought into an operating state and are performed repeatedly in parallel with each other while the system **10** is operating.

[0046] The component mounting control shown in FIG. **7** will be described. As shown in FIG. **7**, steps from step S**102** to step S**114** are performed by the server device **200**. In step S**102**, the stop instructing unit **216** judges whether the state of the stop flag SF is ON. If the state of the stop flag SF is judged to be ON (step S**102**: Yes), the stop instructing unit **216** transmits a stop instruction to the vehicle **100** in step S**114**. Then, step S**102** is performed again.

[0047] If the state of the stop flag SF is judged not to be ON (step S**102**: No), in other words, if the state of the stop flag SF is OFF, the calculating unit **210** acquires detection result from the external sensor **250** in step S**104**. Then, in step S**106**, the calculating unit **210** calculates vehicle positional information using the detection result from the external sensor **250**.

[0048] In step S**108**, the vehicle control commanding unit **212** generates a vehicle control command using the vehicle positional information, and transmits the vehicle control command to the vehicle **100**. Next, step S**102** described above is performed again. In step S**110**, the robot control commanding unit **218** judges whether the vehicle **100** is at a component mounting position set in advance. The "component mounting position" means a position set in advance as a position where the mounting robot **300** is to start mounting of a component on the vehicle **100** in a process of manufacturing the vehicle **100**.

[0049] If the vehicle **100** is judged to be at the component mounting position (step S**110**: Yes), the robot control commanding unit **218** generates a mounting instruction and transmits the mounting instruction to the mounting robot **300** in step S**112**. Then, step S**102** described above is performed again.

[0050] If the vehicle **100** is judged not to be at the component mounting position (step S**110**: No), step S**102** is performed again. In other words, if the vehicle **100** has not yet reached the component mounting position, running control over the vehicle **100** toward the component mounting position is performed continuously. Then, step S**102** described above is performed again.

[0051] The component mounting control shown in FIG. **8** will be described. The control shown in Fig, **8** is performed in parallel with the above-described control shown in FIG. **7**. As shown in FIG. **8**, steps from step S**202** to step S**212** are performed by the server device **200**, and steps from step S**302** to step S**310** are performed by the mounting robot **300**.

[0052] In step S**202** shown in FIG. **8**, the error information acquiring unit **214** judges whether an error signal has been received. If it is judged that an error signal has not been received (step S**202**: No), the error information acquiring unit **214** performs step S**202** repeatedly.

[0053] If it is judged that an error signal has been received (step S**202**: Yes), the stop instructing unit **216** transmits a stop signal to the vehicle **100** and the error information acquiring unit **214** switches the state of the stop flag SF to ON in step S**204**.

[0054] In step S**206**, the robot control commanding unit **218** generates a mounting instruction and transmits the mounting instruction to the mounting robot **300**. Specifically, in the present embodiment, in the event of failure of the on-the-move mounting, the vehicle **100** is stopped and then the robot control commanding unit **218** instructs mounting of a component on the stopped vehicle **100** (hereinafter also called "re-mounting"). Specifically, the mounting instruction in this step corresponds to a "second instruction" of the present disclosure and the robot control commanding unit **218** corresponds to a "second instructing unit" of the present disclosure. In the following description, like the re-mounting control performed in this step, mounting of a component on the stopped vehicle **100** will also be called "stopped mounting."

[0055] In step S**208**, the stop instructing unit **216** judges whether an error signal has been received. If it is judged that an error signal has not been received (step S**208**: No), the vehicle control commanding unit **212** transmits a running restart instruction for instructing restart of running to the vehicle **100** and the error information acquiring unit **214** switches the state of the stop flag SF to OFF in step S**210**.

[0056] If it is judged that an error signal has been received (step S**208**: Yes), the vehicle control commanding unit **212** transmits a stop instruction to the vehicle **100**, in other words, instructs the vehicle **100** to continue the stopped state, and makes announcement to an administrator. The "administrator" mentioned herein is not limited to a person responsible for overall management of the system **10** but includes a worker to conduct restoration work on the occurrence of any abnormality at the system **10** and a worker conducting work near the component mounting step. By continuing the stopped state of the vehicle **100** and making the announcement to the administrator, it becomes possible to suppress prolongation of a situation where abnormality has occurred at the system **10** and manufacture of the vehicle **100** is interrupted.

[0057] The following describes the control by the mounting robot **300**. In step S**302**, the robot controller **315** judges whether a mounting instruction has been received. If it is judged that a mounting instruction has not been received (step S**302**: No), the robot controller **315** performs step S**302** repeatedly.

[0058] If it is judged that a mounting instruction has been received (step S**302**: Yes), the robot controller **315** conducts mounting of a component on the vehicle **100** in step S**304**.

[0059] In step S**306**, the robot controller **315** judges whether mounting error has occurred. If it is judged that mounting error has not occurred (step S**306**: No), in other words, if the on-the-move mounting on the vehicle **100** has been conducted successfully, step S**302** described above is performed again.

[0060] If it is judged that mounting error has occurred (step S**306**: Yes), the robot controller **315** transmits an error signal to the server device **200** in step S**308**.

[0061] In step S**310**, the robot controller **315** retreats the component. More specifically, the robot controller **315** moves the component to a position where the component does not interfere with the vehicle **100**. Then, step S**302** described above is performed again.

[0062] In the system **10** of the above-described embodiment, if a component has not been mounted successfully on the vehicle **100** by the on-the-move mounting, a stop instruction is transmitted to the vehicle **100** and a re-mounting instruction is transmitted to the mounting robot **300**. Thus, even if the component has not been mounted successfully on the vehicle **100** by the on-the-move mounting, it is still possible to attempt re-mounting of the component while a difficulty level in component mounting is reduced by stopping the vehicle **100**. By doing so, it becomes possible to mount the component properly on the vehicle **100**.

[0063] If the component has been mounted successfully by the re-mounting control, running is restarted. This makes it possible to suppress the extension of time required for manufacturing the vehicle **100**, compared to a case where the stopped state is kept continuously.

[0064] The server device **200** includes the robot control commanding unit **218** that generates a motion control signal and transmits the generated motion control signal to the mounting robot **300**. This makes it possible to prevent the control on the side of the mounting robot **300** from growing complicated.

B. Second Embodiment

[0065] FIG. **9** is an explanatory view showing the configuration of a mounting robot **300***a* according to a second embodiment. The mounting robot **300***a* of the second embodiment differs from the mounting robot **300** of the first embodiment in that it includes a processor **311***a* further functioning as a motion control signal generating unit **317** as well as functioning as the robot controller **315**. The other configuration of the mounting robot **300***a* of the second embodiment is the same as that of the mounting robot **300** of the first embodiment. Thus, a common structure will be given the same sign and detailed description thereof will be omitted.

[0066] The motion control signal generating unit **317** receives a mounting instruction from the robot control commanding unit **218** of the server device **200** and generates a motion control signal. In the present embodiment, the mounting instruction received from the robot control commanding unit **218** may be a signal for only instructing the mounting robot **300** to start component mounting control. In response to receipt of this mounting instruction as a trigger, the motion control signal generating unit **317** identifies a position of mounting of a component on the vehicle **100** using detection result from a sensor not shown in the drawings that is a camera belonging to the mounting robot **300***a*, for example, and generates the motion control signal autonomously. In the present embodiment, the robot controller **315** controls each part of the mounting robot **300***a* including the arm part **320** in response to the motion control signal generated by the motion control signal generating unit **317**.

[0067] In system **10** including the mounting robot **300***a* of the second embodiment described above, the mounting robot **300***a* includes the motion control signal generating unit **317**. This allows the mounting robot **300***a* to be controlled autonomously using a motion control signal generated by itself. Thus, it becomes unnecessary for the server device **200** to perform a process for controlling the mounting robot **300***a*, making it possible to prevent the control on the side of the server device **200** from growing complicated.

C. Third Embodiment

[0068] FIG. **10** is a block diagram showing the configuration of a system **10***v* according to a third embodiment. In the present embodiment, the system **10***v* differs from that of the first embodiment in that it does not include the server device **200** and it includes a vehicle control device **110***v* belonging to a vehicle **100***v*. The vehicle **100***v* of the present embodiment is capable of running by autonomous control by the vehicle **100***v*. The other configuration is the same as that of the first embodiment unless specified otherwise.
[0069] In the present embodiment, a processor **111***v* of the vehicle control device **110***v* executes a program PG**1** stored in a memory **112***v*, thereby functioning as a vehicle controller **115***v*, a calculating unit **190**, an error information acquiring unit **192**, a stop instructing unit **194**, and a robot control commanding unit **196**. The vehicle controller **115***v* generates a running control signal using vehicle positional information, and outputs the generated running control signal to operate the actuator group **120**. By doing so, the vehicle **100***v* becomes capable of running by autonomous control. In the present embodiment, memory **112***v* contains a detection model DM, a reference route RR, and a stop flag SF stored in advance in addition to the program PG**1**. The vehicle control device **110***v* of the third embodiment corresponds to the "control device" of the present disclosure. The vehicle controller **115***v* corresponds to the "moving object controller" of the present disclosure.
[0070] FIG. **11** is a flowchart showing a processing procedure for running control of the vehicle **100***v* in the third embodiment. In step S**11**, the processor **111***v* acquires vehicle location information using detection result output from the camera as an external sensor **250**. In step S**12**, the processor **111***v* determines a target location to which the vehicle **100***v* is to move next. In step S**13**, the processor **111***v* generates a running control signal for causing the vehicle **100***v* to run to the determined target location. In step S**14**, the processor **111***v* controls an actuator group **120** using the generated running control signal, thereby causing the vehicle **100***v* to run by following a parameter indicated by the running control signal. The processor **111***v* repeats the acquisition of vehicle location information, the determination of a target location, the generation of a running control signal, and the control over the actuator group **120** in a predetermined cycle. According to the system **10***v* in the present embodiment, it is possible to cause the vehicle **100***v* to run by autonomous control without controlling the vehicle **100***v* remotely using the server **200**.
[0071] Like in the above-described embodiments, even if a component has not been mounted successfully on the vehicle **100***v* by the on-the-move mounting, the system **10***v* of the present embodiment still makes it possible to attempt re-mounting of the component while a difficulty level in component mounting is reduced by stopping the vehicle **100***v*. By doing so, it becomes possible to mount the component properly on the vehicle **100***v*.
D. Other Embodiments
[0072] (D1) In the above-described embodiments, if the state of the stop flag SF is ON, the stop instructing unit **216** transmits a stop instruction for instructing stop, in other words, a stop instruction for instructing to reduce a running speed of the vehicle **100** to **0**. However, the present disclosure is not limited to this. The stop instructing unit **216** may transmit a deceleration instruction for instructing deceleration to an arbitrary speed that is a speed not leading to stop. Specifically, the stop instructing unit **216** generally instructs the vehicle **100** to be decelerated or stopped. The deceleration instruction corresponds to the "first instruction" of the present disclosure. According to this embodiment, if the on-the-move mounting has not been conducted successfully, it is also possible to conduct the on-the-move mounting on the vehicle **100** while the vehicle **100** is at a lower difficulty level in performing control in conjunction with the mounting robot **300** than in the case of the on-the-move mounting on the vehicle **100** running at a normal speed.
[0073] In the embodiment where the deceleration instruction is transmitted instead of the stop instruction, the vehicle control commanding unit **212** may give a running restart instruction for instructing acceleration from a decelerated state to a running speed in normal time in step S**210** of the component mounting control shown in FIG. **8**. The vehicle control commanding unit **212** may give a running restart instruction for instructing acceleration not only to a running speed in normal

time but also to an arbitrary speed higher than a running speed on the occurrence of mounting error. In this embodiment, the vehicle **100** is accelerated if a component has been mounted successfully by the re-mounting control. This makes it possible to suppress the extension of time required for manufacturing the vehicle **100**, compared to a case where the decelerated state is kept continuously.

[0074] (D2) In the above-described embodiments, if the state of the stop flag SF is ON, the stop instructing unit **216** transmits a stop instruction to the vehicle **100**. However, the present disclosure is not limited to this. If the vehicle **100** is configured to be stopped automatically in response to application of a predetermined load or more, for example, the stop instructing unit **216** may not have to transmit a stop instruction to the vehicle **100**. In this embodiment, the server device **200** may not have to include the stop instructing unit **216**. This embodiment also achieves effects comparable to those of the above-described embodiments.

[0075] (D3) In the above-described embodiments, if the state of the stop flag SF is ON, the stop instructing unit **216** transmits an emergency stop signal as a stop instruction. However, the present disclosure is not limited to this. In addition to the emergency stop signal, a running control signal for instructing stop of the vehicle **100** may be transmitted from the vehicle control commanding unit **212**. In this embodiment, the vehicle control commanding unit **212** corresponds to the “first instructing unit” of the present disclosure. This embodiment also achieves effects comparable to those of the above-described embodiments. Furthermore, as the running control signal for instructing stop is transmitted in addition to the emergency stop signal, it is possible to stop the vehicle **100** more reliably. The running control signal for instructing stop of the vehicle **100** may be transmitted instead of the emergency stop signal. In this embodiment, the server device **200** may not have to include the stop instructing unit **216**.

[0076] (D4) In the above-described embodiments, the error information acquiring unit **214** acquires an error signal as mounting error information. However, the present disclosure is not limited to this. The error information acquiring unit **214** may acquire detection result from the external sensor **250** as mounting error information, for example. In this embodiment, the error information acquiring unit **214** may judge whether a component has been mounted successfully on the vehicle **100** by acquiring an image captured by a camera as the external sensor **250** and detecting the positions of the vehicle **100** and the component relative to each other in the captured image, for example. This embodiment also achieves effects comparable to those of the above-described embodiments.

[0077] (D5) In the above-described embodiments, the server device **200** performs control over the vehicle **100** as a component mounting target in response to whether the on-the-move mounting has been conducted successfully. However, the present disclosure is not limited to this. The server device **200** may further perform control over a vehicle other than the vehicle **100** as a component mounting target in response to whether the on-the-move mounting has been conducted successfully. If the on-the-move mounting ends in failure and the vehicle **100** as a component mounting target is stopped, the server device **200** may instruct a different vehicle running behind this vehicle **100** (hereinafter may also be called a “following vehicle”) to be stopped or decelerated in addition to performing step S**204** described above. In this embodiment, it is possible to reduce a likelihood that running of a following moving object will be hindered as a result of shortening of a distance between a moving object on which the on-the-move mounting has not been conducted successfully and the following moving object.

[0078] If a component has not been mounted successfully even by the re-mounting, the server device **200** may instruct the following vehicle to be stopped or decelerated, in other words, may instruct the following vehicle to continue a stopped state or a decelerated state in addition to performing step S**212** described above. In this embodiment, it is possible to restrict running of the vehicle **100** or the following moving object in a situation where the re-mounting has not been conducted successfully, namely, in a situation where some type of abnormality might occur.

[0079] If the on-the-move mounting ends in failure and the vehicle **100** as a component mounting target is stopped, the server device **200** may newly set a component mounting position for conducting the on-the-move mounting on the following vehicle, and a component mounting angle and a component mounting speed for mounting on the following vehicle. These new components mounting position, and component mounting angle and component mounting speed for mounting on the following vehicle may be determined by adding amounts of compensation defined in advance or may be determined using detection result from the external sensor **250**, for example. This embodiment allows a way of the on-the-move mounting on the following vehicle to be changed in response to whether the on-the-move mounting has been conducted successfully on the preceding vehicle, making it possible to reduce a probability of failure of the on-the-move mounting on the following vehicle.

[0080] If the on-the-move mounting ends in failure and the vehicle **100** as a component mounting target is stopped, the server device **200** may conduct the stopped mounting on the following vehicle instead of the on-the-move mounting. In this embodiment, even if the on-the-move mounting ends in failure, it is still possible to attempt re-mounting of a component by the stopped mounting where a difficulty level in component mounting is lower than in the on-the-move mounting. This allows the component to be mounted properly on the vehicle **100**.

[0081] (D6) In the above-described embodiments, the motion control signal generating unit **317** receives a mounting instruction from the robot control commanding unit **218** of the server device **200** and generates a motion control signal. However, the present disclosure is not limited to this. The motion control signal generating unit **317** may generate a motion control signal if the vehicle **100** is determined to be at a component mounting position using detection result from a sensor that may be a camera belonging to the mounting robot **300***a*, for example, without receiving a mounting instruction from the robot control commanding unit **218**. Specifically, the mounting robot **300** may realize motion of mounting a component on the vehicle **100** independently of an instruction from a different device. This embodiment also achieves effects comparable to those of the above-described embodiments.

[0082] (D7) In each of the above-described embodiments, the external sensor **250** is not limited to the camera but may be the distance measuring device, for example. The distance measuring device is a light detection and ranging (LiDAR) device, for example. In this case, detection result output from the external sensor **250** may be three-dimensional point cloud data representing the vehicle **100**. The server **200** and the vehicle **100** may acquire the vehicle location information through template matching using the three-dimensional point cloud data as the detection result and reference point cloud data, for example.

[0083] (D8) In the above-described first embodiment, the server **200** performs the processing from acquisition of vehicle location information to generation of a running control signal. By contrast, the vehicle **100** may perform at least part of the processing from acquisition of vehicle location information to generation of a running control signal. For example, embodiments **(1)** to **(3)** described below are applicable, for example.

[0084] (1) The server **200** may acquire vehicle location information, determine a target location to which the vehicle **100** is to move next, and generate a route from a current location of the vehicle **100** indicated by the acquired vehicle location information to the target location. The server **200** may generate a route to the target location between the current location and a destination or generate a route to the destination. The server **200** may transmit the generated route to the vehicle **100**. The vehicle **100** may generate a running control signal in such a manner as to cause the vehicle **100** to run along the route received from the server **200** and control an actuator using the generated running control signal.

[0085] (2) The server **200** may acquire vehicle location information and transmit the acquired vehicle location information to the vehicle **100**. The vehicle **100** may determine a target location to which the vehicle **100** is to move next, generate a route from a current location of the vehicle **100** indicated by the received vehicle location information to the target location, generate a running control signal in such a manner as to cause the vehicle **100** to run along the generated route, and control an actuator using the generated running control signal.

[0086] (3) In the foregoing embodiments (1) and (2), an internal sensor may be mounted on the vehicle **100**, and detection result output from the internal sensor may be used in at least one of the generation of the route and the generation of the running control signal. The internal sensor is a sensor mounted on the vehicle **100**. The internal sensor includes, for example, a sensor that detects the state of motion of the vehicle **100**, the state of operation of various parts of the vehicle **100**, and the environment surrounding the vehicle **100**. More specifically, the internal sensor might include a camera, LiDAR, a millimeter wave radar, an ultrasonic wave sensor, a GPS sensor, an acceleration sensor, and a gyroscopic sensor, for example. For example, in the foregoing embodiment (1), the server **200** may acquire detection result from the internal sensor, and in generating the route, may reflect the detection result from the internal sensor in the route. In the foregoing embodiment (1), the vehicle **100** may acquire detection result from the internal sensor, and in generating the running control signal, may reflect the detection result from the internal sensor in the running control signal. In the foregoing embodiment (2), the vehicle **100** may acquire detection result from the internal sensor, and in generating the route, may reflect the detection result from the internal sensor in the route. In the foregoing embodiment (2), the vehicle **100** may acquire detection result from the internal sensor, and in generating the running control signal, may reflect the detection result from the internal sensor in the running control signal.

[0087] (D9) In the third embodiment, the vehicle **100***v* may be equipped with an internal sensor, and detection result output from the internal sensor may be used in at least one of generation of a route and generation of a running control signal. For example, the vehicle **100***v* may acquire detection result from the internal sensor, and in generating the route, may reflect the detection result from the internal sensor in the route. The vehicle **100***v* may acquire detection result from the internal sensor, and in generating the running control signal, may reflect the detection result from the internal sensor in the running control signal.

[0088] (D10) In the above-described embodiment in which the vehicle **100** can be running by autonomous control, the vehicle **100** acquires vehicle location information using detection result from the external sensor. By contrast, the vehicle **100** may be equipped with an internal sensor, the vehicle **100** may acquire vehicle location information using detection result from the internal sensor, determine a target location to which the vehicle **100** is to move next, generate a route from a current location of the vehicle **100** indicated by the acquired vehicle location information to the target location, generate a running control signal for running along the generated route, and control an actuator of the vehicle **100** using the generated running control signal. In this case, the vehicle **100** is capable of running without using any detection result from an external sensor. The vehicle **100** may acquire target arrival time or traffic congestion information from outside the vehicle **100** and reflect the target arrival time or traffic congestion information in at least one of the route and the running control signal. The functional configuration of the system **10** may be entirely provided at the vehicle **100**. Specifically, the processes realized by the system **10** in the present disclosure may be realized by the vehicle **100** alone. In this embodiment, it is judged whether a component has been mounted successfully on the vehicle **100** as follows. Using a status of connection between the component and a vehicle-mounted LAN, for example, if connection with a predetermined component has not been established at given timing, it is possible to determine that the on-the-move mounting of this component has not been conducted successfully.

[0089] (D11) In the above-described first embodiment, the server **200** automatically generates a running control signal to be transmitted to the vehicle **100**. By contrast, the server **200** may generate a running control signal to be transmitted to the vehicle **100** in response to operation by an external operator existing outside the vehicle **100**. For example, the external operator may operate an operating device including a display on which a captured image output from the external sensor **250** is displayed, steering, an accelerator pedal, and a brake pedal for operating the vehicle **100** remotely, and a communication device for making communication with the server **200** through wire communication or wireless communication, for example, and the server **200** may generate a running control signal responsive to the operation on the operating device.

[0090] (D12) The vehicle **100** may be manufactured by combining a plurality of modules. The module means a unit composed of one or more components grouped according to a configuration or function of the vehicle **100**. For example, a platform of the vehicle **100** may be manufactured by combining a front module, a center module and a rear module. The front module constitutes a front part of the platform, the center module constitutes a center part of the platform, and the rear module constitutes a rear part of the platform. The number of the modules constituting the platform is not limited to three but may be equal to or less than two, or equal to or greater than four. In addition to or instead of the platform, any parts of the vehicle **100** different from the platform may be modularized. Various modules may include an arbitrary exterior component such as a bumper or a grill, or an arbitrary interior component such as a seat or a console. Not only the vehicle **100** but also any types of moving object may be manufactured by combining a plurality of modules. Such a module may be manufactured by joining a plurality of components by welding or using a fixture, for example, or may be manufactured by forming at least part of the module integrally as a single component by casting. A process of forming at least part of a module as a single component is also called Giga-casting or Mega-casting. Giga-casting can form each part conventionally formed by joining multiple parts in a moving object as a single component. The front module, the center module, or the rear module described above may be manufactured using Giga-casting, for example.

[0091] (D13) Conveying the vehicle **100** using running of the vehicle **100** by unmanned driving is also called “self-running conveyance.” A configuration for realizing self-running conveyance is also called a “Remote Control auto Driving system. Producing the vehicle **100** using self-running conveyance is also called “self-running production.” In self-running production, for example, at least part of the conveyance of the vehicle **100** is realized by self-running conveyance in the factory KJ where the vehicle **100** is manufactured.

[0092] (D14) A configuration for realizing running of the vehicle **100** by unmanned driving is also called a “Remote Control auto Driving system”. Conveying the vehicle **100** using Remote Control Auto Driving system is also called “self-running conveyance”. Producing the vehicle **100** using self-running conveyance is also called “self-running production”. In self-running production, for example, at least part of the conveyance of the vehicle **100** is realized by self-running conveyance in a factory where the vehicle is manufactured.

[0093] The present disclosure is not limited to the above-described embodiments but is feasible in various configurations within a range not deviating from the purport of the disclosure. For example, technical features in the embodiments corresponding to those in each of the aspects described in SUMMARY may be replaced or combined, where appropriate, with the intention of solving some or all of the aforementioned problems or achieving some or all of the aforementioned effects. Unless being described as absolute necessities in the present specification, these technical features may be deleted, where appropriate. The present disclosure may be realized in the following aspects, for example.

[0094] (1) According to one aspect of the present disclosure, a control device is provided that controls mounting of a component on a moving object in the process of manufacturing the moving object. The control device comprises: an information acquiring unit that acquires first information that is information relating to unsuccessful mounting of the component by a mounting device to conduct on-the-move mounting that is mounting of the component on the moving object while the moving object is moving by unmanned driving; a first instructing unit that gives a first instruction for instructing deceleration or stop to the moving object if the first information is acquired; and a second instructing unit that gives a second instruction for instructing re-mounting to the mounting device after transmission of the first instruction. The re-mounting is mounting of the component on the moving object while the moving object is decelerated or stopped.

[0095] According to the control device of this aspect, if the component has not been mounted successfully on the moving object by the on-the-move mounting, the first instruction for instructing deceleration or stop is transmitted to the moving object, and the second instruction for instructing the re-mounting is transmitted to the mounting device. Thus, even if the component has not been mounted successfully on the moving object by the on-the-move mounting, it is still possible to attempt the re-mounting of the component while a difficulty level in component mounting is reduced by decelerating or stopping the moving object. By doing so, it becomes possible to mount the component properly on the moving object.

[0096] (2) In the above-described aspect, the first instructing unit may further instruct a following moving object to be decelerated or stopped if the first information is acquired. The following moving object is a different moving object moving behind the moving object.

[0097] According to the control device of this aspect, if the on-the-move mounting has not been conducted successfully, the following moving object is instructed to be decelerated or stopped. This makes it possible to suppress shortening of a distance between the moving object on which the on-the-move mounting has not been conducted successfully and the following moving object.

[0098] (3) In the above-described aspect, if the component has been mounted successfully on the moving object by the re-mounting, the control device may further instruct the moving object to be accelerated or restart moving.

[0099] According to the control device of this aspect, moving is restarted or accelerated if the component has been mounted successfully by the re-mounting control. This makes it possible to suppress extension of time required for manufacturing the moving object, compared to a case where the decelerated or stopped state is kept continuously.

[0100] (4) In the above-described aspect, if the component has not been mounted successfully on the moving object by the re-mounting, the first instructing unit may further perform at least one of: instructing the moving object to continue a decelerated or stopped state; and instructing a following moving object to be decelerated or stopped. The following moving object is a different moving object moving behind the moving object.

[0101] According to the control device of this aspect, if the component has not been mounted successfully on the moving object by the re-mounting, at least one of instructing the moving object to continue a decelerated or stopped state and instructing a following moving object to be decelerated or stopped is performed further. The following moving object is a different moving object moving behind the moving object. Thus, it is possible to restrict moving of the moving object or the following moving object in a situation where the re-mounting has not been conducted successfully, namely, in a situation where some type of abnormality might occur.

[0102] (5) In the above-described aspect, the control device may further comprise a motion control signal generating unit that generates a motion control signal for causing the mounting device to make motion, and the second instructing unit may transmit the motion control signal as the second instruction to the mounting device.

[0103] According to the control device of this aspect, as the control device generates the motion control signal and transmits the motion control signal to the mounting device, it is possible to prevent the control on the side of the mounting device from growing complicated.

[0104] (6) According to another aspect of the present disclosure, a system is provided. The system comprises: the control device described in any one of the first to fourth aspects; and the mounting device. The mounting device includes: a receiving unit that receives the second instruction; a motion control signal generating unit that receives the second instruction and generates a motion control signal for causing the mounting device to make motion; and a device controller that controls the motion of the mounting device in response to the motion control signal.

[0105] According to the system of this aspect, as the mounting device generates the motion control signal, it is possible to prevent the control on the side of the control device from growing complicated.

[0106] (7) According to another aspect of the present disclosure, a mounting device that mounts a

component on a moving object is provided. The mounting device comprises a device controller that conducts re-mounting if the component has not been mounted successfully on the moving object while the moving object is moving by unmanned driving. The re-mounting is mounting of the component on the moving object while the moving object is decelerated or stopped.
[0107] According to the mounting device of this aspect, even if the component has not been mounted successfully on the moving object by the on-the-move mounting, it is still possible to attempt the re-mounting of the component on the moving object while the moving object is at a difficulty level in component mounting reduced by decelerating or stopping the moving object. By doing so, it becomes possible to mount the component properly on the moving object.
[0108] (8) In the above-described aspect, the mounting device may comprise a motion control signal generating unit that generates a motion control signal for causing the mounting device to make motion, and the device controller may conduct the re-mounting in response to the motion control signal generated by the motion control signal generating unit.
[0109] According to the mounting device of this aspect, as the mounting device generates the motion control signal, it is possible to prevent the control on the side of the control device from growing complicated.
[0110] (9) According to another aspect of the present disclosure, a moving object is provided that is movable by unmanned driving in a process of manufacturing the moving object. The moving object comprises a moving object controller that performs: deceleration or stopping if on-the-move mounting has not been conducted successfully, the on-the-move mounting being mounting of a component on the moving object while the moving object is moving by the unmanned driving; and acceleration or restart of moving if the component has been mounted successfully on the moving object by re-mounting that is mounting of the component on the moving object while the moving object is decelerated or stopped.
[0111] According to the moving object of this aspect, even if the component has not been mounted successfully on the moving object by the on-the-move mounting, it is still possible to attempt the re-mounting of the component while a difficulty level in component mounting is reduced by the deceleration or stopping. By doing so, it becomes possible to mount the component properly on the moving object.

Claims

1. A control device that controls mounting of a component on a moving object in a process of manufacturing the moving object, the control device comprising: an information acquiring unit that acquires first information that is information relating to unsuccessful mounting of the component by a mounting device to conduct on-the-move mounting that is mounting of the component on the moving object while the moving object is moving by unmanned driving; a first instructing unit that gives a first instruction for instructing deceleration or stop to the moving object if the first information is acquired; and a second instructing unit that gives a second instruction for instructing re-mounting to the mounting device after transmission of the first instruction, the re-mounting being mounting of the component on the moving object while the moving object is decelerated or stopped.
2. The control device according to claim 1, wherein the first instructing unit further instructs a following moving object to be decelerated or stopped if the first information is acquired, the following moving object being a different moving object moving behind the moving object.
3. The control device according to claim 1, wherein if the component has been mounted successfully on the moving object by the re-mounting, the control device further instructs the moving object to be accelerated or restart moving.
4. The control device according to claim 1, wherein if the component has not been mounted successfully on the moving object by the re-mounting, the first instructing unit further performs at

least one of: instructing the moving object to continue a decelerated or stopped state; and instructing a following moving object to be decelerated or stopped, the following moving object being a different moving object moving behind the moving object.

5. The control device according to claim 1, further comprising: a motion control signal generating unit that generates a motion control signal for causing the mounting device to make motion, wherein the second instructing unit transmits the motion control signal as the second instruction to the mounting device.

6. The control device according to claim 4, further comprising: a motion control signal generating unit that generates a motion control signal for causing the mounting device to make motion, wherein the second instructing unit transmits the motion control signal as the second instruction to the mounting device.

7. A system comprising: the control device according to claim 1; and the mounting device, wherein the mounting device includes: a receiving unit that receives the second instruction; a motion control signal generating unit that receives the second instruction and generates a motion control signal for causing the mounting device to make motion; and a device controller that controls the motion of the mounting device in response to the motion control signal.

8. A system comprising: the control device according to claim 4; and the mounting device, wherein the mounting device includes: a receiving unit that receives the second instruction; a motion control signal generating unit that receives the second instruction and generates a motion control signal for causing the mounting device to make motion; and a device controller that controls the motion of the mounting device in response to the motion control signal.

9. A mounting device that mounts a component on a moving object, comprising: a device controller that conducts re-mounting if the component has not been mounted successfully on the moving object while the moving object is moving by unmanned driving, the re-mounting being mounting of the component on the moving object while the moving object is decelerated or stopped.

10. The mounting device according to claim 9, comprising: a motion control signal generating unit that generates a motion control signal for causing the mounting device to make motion, wherein the device controller conducts the re-mounting in response to the motion control signal generated by the motion control signal generating unit.

11. A moving object movable by unmanned driving in a process of manufacturing the moving object, the moving object comprising: a moving object controller that performs: deceleration or stopping if on-the-move mounting has not been conducted successfully, the on-the-move mounting being mounting of a component on the moving object while the moving object is moving by the unmanned driving; and acceleration or restart of moving if the component has been mounted successfully on the moving object by re-mounting that is mounting of the component on the moving object while the moving object is decelerated or stopped.